

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1538

Steamboat 138 kV
March 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1538. The Interconnection Customer proposes a 131.1 MW battery energy storage facility interconnecting at the Steamboat 138 kV substation in the Virginia area. The request was received on August 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 477 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,985,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Huntington Court Substation to Steamboat 138 kV
Contingency: Vinton Substation to Vc Porterdale Hydroelectric 138 kV
Loading: 108.7% (Rating: 224 MVA)
DFAX: 51.4%
Pre-project loading: 78.7%
2. Facility: Muddy Ranch to Steamboat 138 kV
Contingency: Huntington Court to Steamboat 138 kV
Loading: 108.6% (Rating: 184 MVA)
DFAX: 100.0%
Pre-project loading: 37.2%
3. Facility: Louisa Generation Facility to North Anna 345 kV line
Contingency: Louisa Generation Facility to Carrabelle Substation 345 kV Line
Loading: 101.6% (Rating: 2,511 MVA)
DFAX: 11.8%
Pre-project loading: 101.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Huntington Court Substation to Steamboat 138 kV	Reconductor line	\$1,458,000	37% (\$534,000)
Muddy Ranch to Steamboat 138 kV	Reconductor line	\$1,063,000	100% (\$1,063,000)
Louisa Generation Facility to North Anna 345 kV	Reconductor line	\$29,560,000	1% (\$388,000)
	Total allocated to project		\$1,985,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.